

JNTU Hyderabad

MID-TERM EXAMINATION ? 2025

Earthquake Resistant Design of Structures ? Mid-Term Examination, 2025

Subject Code: CE602 Semester: 7 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Earthquake Basics and Seismology)

1. Compare and contrast the key concepts involved in Earthquake Basics and Seismology. [12 Marks]
2. Write short notes on Earthquake Basics and Seismology. [12 Marks]
3. Explain Earthquake Basics and Seismology with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION B (Unit 2: Response Spectrum Analysis)

4. Define Response Spectrum Analysis. Explain its significance in Earthquake Resistant Design of Structures with a suitable example. [8 Marks]
5. How is Response Spectrum Analysis implemented? Give an algorithm / procedure and explain. [12 Marks]
6. Discuss the engineering applications of Response Spectrum Analysis. [8 Marks]

SECTION C (Unit 3: Ductile Detailing of RC)

7. Define Ductile Detailing of RC. Explain its significance in Earthquake Resistant Design of Structures with a suitable example. [12 Marks]
8. Define Ductile Detailing of RC. Explain its significance in Earthquake Resistant Design of Structures with a suitable example. [12 Marks]
9. Discuss the engineering applications of Ductile Detailing of RC. [12 Marks]

SECTION D (Unit 4: Masonry Structures)

10. Write short notes on Masonry Structures. [12 Marks]
11. Solve a problem related to Masonry Structures. Show all the steps clearly. [8 Marks]
12. Compare and contrast the key concepts involved in Masonry Structures. [12 Marks]

SECTION E (Unit 5: Base Isolation and Control)

13. Write short notes on Base Isolation and Control. [12 Marks]
14. Describe the working principle of Base Isolation and Control with necessary equations. [12 Marks]
15. Discuss the engineering applications of Base Isolation and Control. [12 Marks]

***** End of Question Paper *****